

Dominic Duxbury, PhD – Data Consultant

Data Consultant with experience in research and as a data engineer working on real-time systems.

Experience and Education Timeline

Senior Data Consultant | Independent (Contract)

February 2025 – April 2025 | Department of Health and Social Care (DHSC)

For this three month discovery project I analysed ways forward for the Adult Social Care Workforce Data Set (ASC-WDS) and workforce data as a whole. I worked alongside a technical architect, splitting responsibilities between data and system processes. We carried out discovery interviews together and drew up diagrams detailing the current system, moving on to create a vision of a future system, running on the DHSC in-house cloud within **Azure**.

Senior Data Consultant | Hippo Digital (Contract)

December 2024 – August 2025 | Pathogena EIT

In this role I gained experience productionising cutting-edge science from a group of Oxford scientists. The stack consisted of a **django** app using a **ReactJS frontend** to pass DNA data files through to a **Kubernetes** cluster of data processors on **Oracle Cloud**. I contributed towards the release of a pipeline for the automatic characterisation of new variants of SARS-CoV-2 plus I designed and implemented a new logging pipeline for business KPIs with **structlog** and **Loki**. My role included scoping pieces of work, ticketing and other facets of managing a small team of engineers.

April 2024 – December 2024 | Flutter Int.

The Flutter cloud estate uses a patchwork of cloud technologies due to repeat acquisitions. The databases were judged with regard to Flutter's data retention policy, to ensure GDPR is followed and reduce waste. Over 7 TB of data was highlighted (mostly container image backups) to be deleted of the 15+ TB surveyed. My role included developing software to automate the extraction and analysis of data plus documentation of acquired data and decisions made.

February 2024 – April 2024 | Department of Health and Social Care (DHSC)

Subcontracted at the Department of Health and Social Care alongside a Hippo team as part of a departmental data migration into **Azure**. We had 8 weeks to discover and document 5 systems, then present a migration plan for the products. My role included interviewing product owners, lending data engineering knowledge to the creation of new architectures and applying options analysis techniques from my PhD to highlight trade-offs between different approaches.

Senior Data Engineer | Netacea

February 2022 – December 2023

I have led the development of a new platform for delivering real-time models at Netacea. This platform has leveraged **pyFlink** to create a new modular strategy for delivering data products that can scale up to billions of requests per day. Leadership experience included creating tickets, planning sprints and leading meetings to prioritise work for a team of six. Developer experience included: applying **python** and **docker** to create a streamlined developer experience with the highlight being a containerised locally runnable **Kafka/pyFlink** environment for integration tests.

PhD, Computer Science | University of Manchester and BAE Systems

October 2017 – October 2021

As part of an industrial PhD (iCASE) studentship, I applied within a team (at BAE Systems DI) and investigated (at University of Manchester) methodologies for combining stream processing engines decision support algorithms to improve the effectiveness of dynamic decision support.

- A cloud architecture including a **Flink** cluster on **Google Cloud Platform** was used to deliver a user experiment during coronavirus restrictions. This architecture enabled novel decision making algorithms to be executed in real-time.
- A headless version of this software was designed to provide comparison between methods (e.g. **TOPSIS** or **AHP**) embedded within the broader genetic algorithm as kernels for decision support.

Data Scientist | Laterooms.com

June 2015 – July 2019

I joined Laterooms as a 3-month industrial placement then moved to a permanent role. I worked with **Google Cloud Platform** including **BigQuery** for big data analysis and infrastructure teams to restructure and centralise event streams and databases using **Kafka** and **Docker**.

BSC, Computer Science and Mathematics | University of Manchester

October 2013 – June 2017 | **BA (Hons) 1st Class.**

For my mathematics options I focused on statistics and pure mathematics, including group theory and information theory. Computer Science included a project on sentiment analysis.

A Levels | Cardinal Newman College, Preston | 2011 – 2013

AAA in Maths, Further Maths and Chemistry, **B (79%)** in AS Level Economics.

GCSE | Holy Cross Catholic College, Chorley | 2006 – 2011

6 A*, including 3 A* in sciences and an A* in maths, **5 A**, including English.

Publications

Dynamic Decision Making for Situational Awareness using Drones

Expert Systems with Applications – Volume 252, Part B, 15 October 2024, 124057

<https://www.sciencedirect.com/science/article/abs/pii/S0957417424009230>

This paper presents desiderata for dynamic decision support and compares kernel methods of a genetic algorithm with regards to the consistency of trade-offs between criteria and the stability of results under changes to criteria values.

Talks

- | | |
|-----------|---|
| May. 2025 | "How to handle homelab paranoia with Arroyo", at PyData MCR. |
| Mar. 2023 | "Tracking the Journey: Exploring Data Provenance in Machine Learning", at PyData MCR. |
| Apr. 2020 | "Decision Support for Transport", at Data Science in Transport Conference. |
| Jun. 2018 | "Trusted and Auditable Decision Aids over Data Streams", at Combat Systems Expo 2018. |